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Service life evaluation of marine concrete structures considering spatial and temporal characteristics: A framework based on multi training-MCS-NLS

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ABSTRACT

Corrosion of steel caused by chloride corrosion is the main factor leading to the service life of marine structures shorter than the design life. However, this failure mechanism is affected by the coupling of multiple factors such as material composition, environmental parameters and service time, which cannot be adequately revealed by traditional models. Based on this, this paper proposed a framework to assess the service life of marine concrete structure. Firstly, a database containing 711 sets of concrete surface chloride ion concentrations and diffusion coefficients in different exposure zones (splash, tidal and submerged zone) was constructed. Tree models were used to establish multi-training models to predict the above indicators. The improved service life evaluation model was established by considering multi-parameters. The influence of uncertain parameters on structural failure probability was quantified by MCS. Furthermore, the NLS-LM model was used to optimize the cover thickness under multi-parameter combinations. The results show that random forest exhibits the highest prediction accuracy, with mean prediction values of 1.07,1.11 and 1.13 in splash, tidal and submerged zone, respectively. Evaluation model based on multi training and MCS can effectively consider the influence of spatial and temporal characteristics, material composition on the service life of marine structure. The probability distribution and combinations of uncertain parameters will significantly affect the failure probability. In addition, the optimization model based on NLS-LM can effectively guide the design of cover thickness under any marine environment.

1. Introduction

The service life of marine concrete structures will be much less than the design service life, which is affected by wave erosion and the diffusion of various harmful ions. This has consequently prompted a focus on the durability and service life design of such structures in relevant research [1–4]. Existing research indicates that steel corrosion has become a key factor affecting the service life of structures [5–8]. Specifically, chloride ions will reach the surface of the reinforcing steel through diffusion and other reactions, thereby disrupting the passivation film (Fig. 1). When the concentration of chloride ions on the steel surface exceeds the critical chloride ion concentration c_r, the steel will rust and lead to structural damage [9–12].

Existing codes such as EC2 [13] and DuraCrete [14] recommend semi-probability models, which establish the probability models considering the chloride ion transport coefficient with a constant value. So these methods cannot effectively consider the influence of the time-dependent behavior of concrete on the service life of structures. Based on the reliability method, scholars have proposed a series of life assessment models [15–17]. Yu et al. [18] established a life assessment model for coral aggregate concrete structures by analyzing the degradation behavior of chloride ion diffusion coefficients over exposure time. Guo et al. [19] studied the impact of chloride ion transport models and environmental types on the service life of structures by establishing a dynamic Bayesian framework. Xia et al. [20] established an evaluation framework for the reliability of corroded concrete considering uncertain parameters such as coverage depth and surface chloride concentration.

The transport mechanisms of chloride ions vary in different zones of marine structures (the atmospheric, splash, tidal, and submerged zone) [21–23]. In the submerged zone, chloride ions experience pure diffusion movement, as the concrete can be assumed to be in a saturated state [24–26]. In the splash zone and tidal zone, absorption is another mechanism of chloride transport [27]. In the atmospheric zone, the chloride ion transport mechanism is more complex, as is affected by many factors such as wind speed, rainfall and coastal distance [28,29].

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Nomenclature

A_c	Coefficient related to the marine environment
C_s	Chloride ion concentration on the surface of concrete
C_b	Chloride ion concentration related to coastal conditions
Cl^- in sea	Seawater chloride ion concentration
D	Diffusion coefficient
$f_i(c)$	Experimental value for the individual sample
$f(c)_{mean}$	Mean experimental value for the sample set
FA	Fly ash content
GEP	Gene expression programming
GBDT	Gradient boosting decision tree
MCS	Monte carlo simulation
MAE	Mean absolute error
NLS	Nonlinear least squares
OPC	Cement content
P_f	Failure probability
q_i	Predicted value of the individual sample
q_{mean}	Mean predicted value for the sample set
r	Replacement ratio of recycled coarse aggregate
R	Chloride binding capacity
RF	Random forest
R^2	Coefficient of determination
RMSE	Root mean squared error
SG	Slag content
SF	Silica fume content
t	Exposure time
T	Seawater temperature
w/b	Water-to-binder ratio
K	Deterioration coefficient
XGB	XGBoost
x_k	Cover thickness

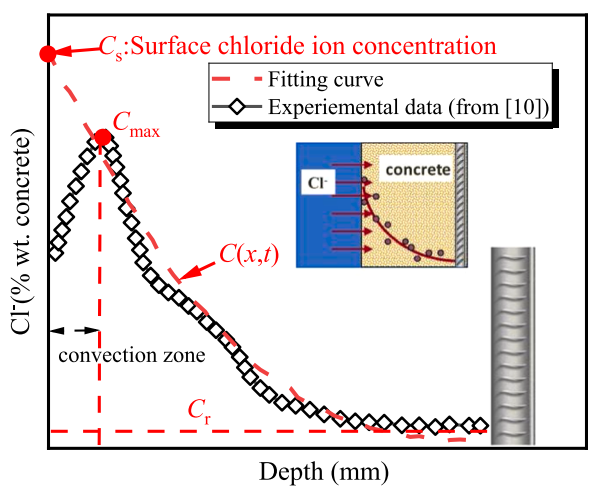

Fig. 1. Transport process of chloride ion in concrete.

Scholars have conducted extensive research on chloride ion transport methods in different zones [30,31]. Yang et al. [32,33] proposed a method for calculating the surface chloride ion concentration of concrete in the atmospheric zone and the submerged zone, which takes into account the material composition and some environmental factors. Based on Fick 's second law, Xu et al. [34] proposed a method for calculating the surface concentration of chloride ions in tidal zone.

Relevant reports have confirmed that the time-dependent behavior of chloride ion diffusion coefficient has a significant effect on the diffusion of chloride ions in concrete. Justnes et al. [35] proposed a chloride diffusion method considering chloride concentration and exposure time. Zhang et al. [36] established a time-dependent model of chloride diffusion coefficient in concrete, thereby quantifying the influence of water-cement ratio, admixtures, and exposure time.

In summary, existing methods for chloride diffusion assessment are based on single-area field tests with small sample characteristics, which cannot effectively analyze the spatiotemporal features of different areas. Moreover, formulas that only consider partial parameter regression cannot effectively evaluate the service life of marine concrete structures under high-dimensional index sets [37,38].

Machine learning is widely used in the performance evaluation of concrete structures, due to its strong nonlinear processing and multidimensional set processing capabilities [39–43]. Guo et al. [44] combined random forest and fuzzy inference methods to predict the chloride ion concentration on the surface of concrete. Taffese et al. [45] established an interpretable XGB model to analyze non-steady chloride migration in concrete. Ashrafian et al. [46] utilized the GEP to quantify the influence of factors such as water-cement ratio and exposure time on chloride diffusion.

Considering the multi-dimensional and uncertain characteristics of the parameters affecting the service life of marine concrete structures, it will be difficult for ML models to train the law of such complex probability events [47–49]. Based on this, this paper first combines ML algorithms and probability method to establish a framework for marine concrete life evaluation, and then quantifies the influence of uncertain parameters and explains the degradation mechanism of such structures (Fig. 2).

The framework can be divided into the following parts: (1) The measured data of 11 ports were collected and a database containing 711 samples was created. Tree models (RF, GBDT and XGB) were used to establish a multi-training model to predict the chloride ion concentration on concrete surfaces and chloride ion diffusion coefficients. It should be noted that the input indicators of the model include environmental parameters (seawater chloride ion concentration, temperature), material components (cement, water, fly ash, slag and silica fume) and exposure time, which indicated that the index set can effectively evaluate the influence of spatial and temporal characteristics and material components in different regions; (2) The prediction results of the multi training model were used as input indicators and the effects of parameters such as temperature, chloride binding capacity, and degradation coefficient were considered. Based on these, the improved chloride ion transport model and service life evaluation model were constructed. Monte Carlo simulation (MCS) was used to quantify the influence of uncertain parameters on structural failure probability; (3) The nonlinear least squares (NLS) combined with Levenberg-Marquardt (LM) algorithm was used to optimize the cover thickness of concrete under multi-parameter combination, which provides a basis for engineering application.

2. Machine learning algorithms

2.1. Random forest

RF is a robust model of ensemble learning, frequently employed in addressing regression issues [50]. This algorithm forms multiple decision trees by randomly selecting subsets of features and data, and then computes the value average to determine the final prediction outcome, which significantly diminishes the model's overfitting risk and boosts its generalization ability [51,52].

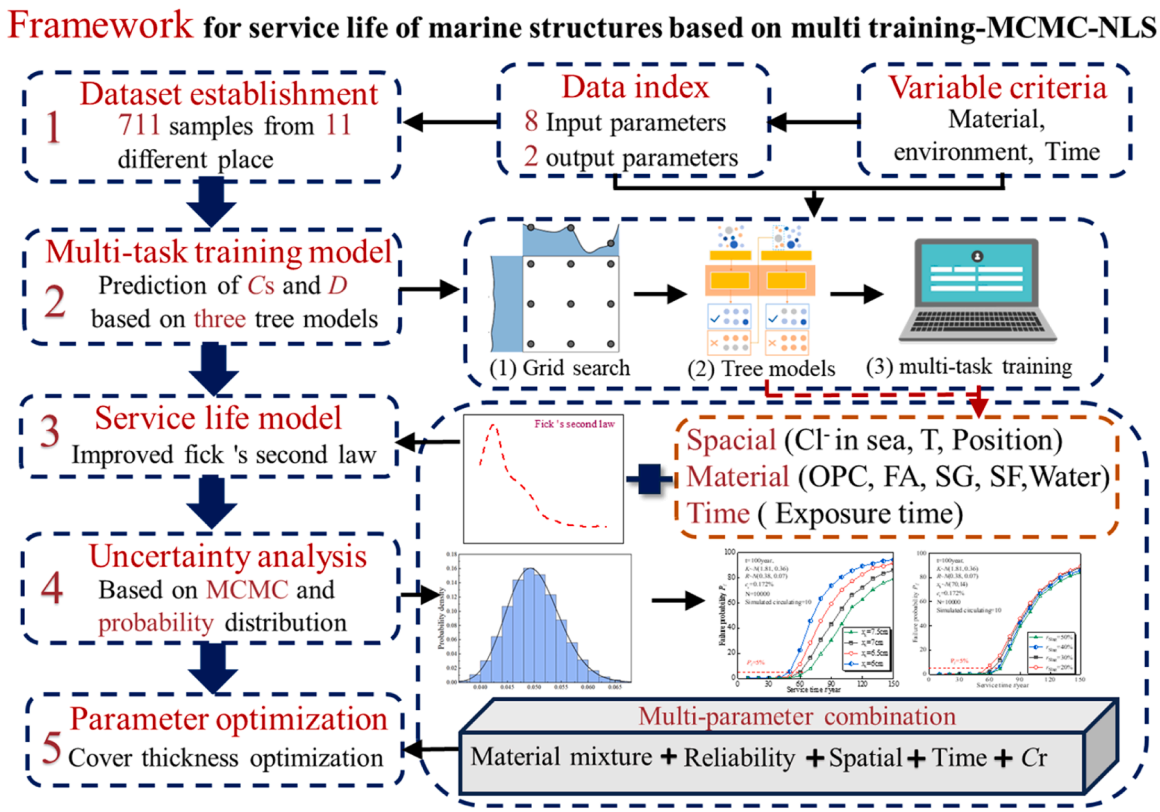

Fig. 2. Service life evaluation model framework for marine concrete structures.

Table 1
Algorithm mechanism and advantages and disadvantages.

	RF	GBDT	XGB
Mechanism	Bagging	Boosting	Boosting
Advantages	Strong interpretability, high degree of nonlinearity	Good robustness, good accuracy of fitting residuals	Regularization, strong explanatory
Disadvantages	Large computational cost, high complexity	Easy to overfit, strong dependence on data distribution	High tuning parameters

2.2. Gradient boosting decision tree

GBDT is an ensemble learning model based on decision trees, which enhances the prediction of regression models through optimization of the loss function [53,54]. The model fits the residuals using the gradient descent method to ensure the model's predictive loss is minimized in the direction of the negative gradient. During each iteration, the model will adjust the weights of new trees for ongoing optimization [55].

2.3. Extreme gradient boosting

XGB enhances the model's predictive performance by iterating multiple decision tree models. The algorithm uses taylor expansion and second-order derivative calculations to optimize the loss function [56], which amplifies XGB's capacity to manage extensive datasets. Compared with GBDT, XGB adds regularization processing. However, although regularization alleviates the over-fitting problem of the model to a certain extent, it will lead to a decrease in the complexity of the model in the early stage of iteration, thereby reducing the prediction accuracy of the model. The comparison of the three algorithm mechanisms and their advantages and disadvantages can be seen in Table 1.

2.4. Performance metrics

The study used the coefficient of determination (R^2), mean absolute error (*MAE*), and root mean squared error (*RMSE*) to evaluate the predictive performance of the multi-training model. It is noteworthy that using statistical indicators to gauge the prediction of ML models significantly enhances the resistance to data noise. The calculation methods for different indicators are as follows:

$$R^2 = \left(\frac{\sum_{i=1}^{n} (q_i - q_{mean})(f_i(c) - f(c)_{mean})}{\sqrt{\sum_{i=1}^{n} (q_i - q_{mean})^2 \sum_{i=1}^{n} (f_i(c) - f(c)_{mean})^2}} \right)^2 \tag{1}$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (q_i - f_i(c))^2} \tag{2}$$

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |q_i - f_i(c)| \tag{3}$$

Where q_i is the predicted value for the individual sample, q_{mean} is the mean predicted value for the sample set, $f_i(c)$ is the experimental value for the individual sample, and $f(c)_{mean}$ is the mean experimental value for the sample set.

3. Database description and analysis

3.1. Collection criteria for variables and database

The correlation between variables has a significant impact on the prediction results of the model. Therefore, formulating reasonable principles for variable and data selection will be advantageous in

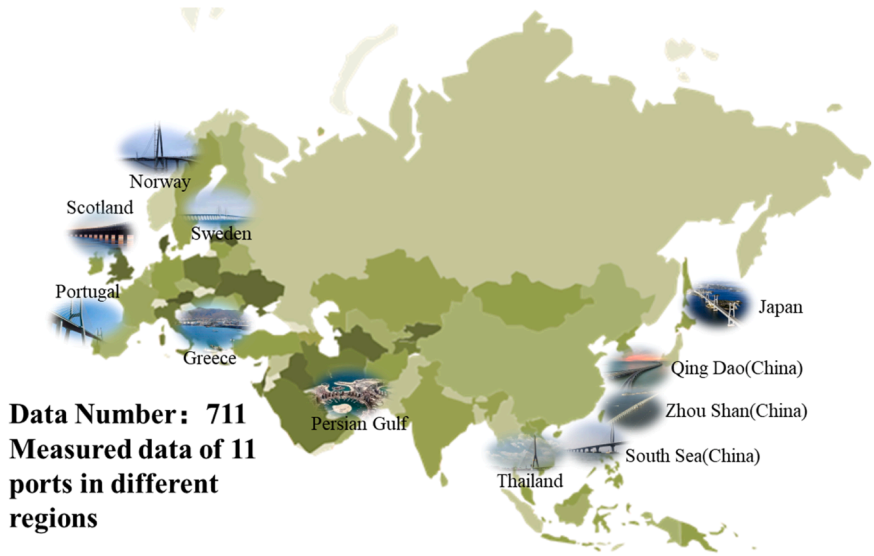

Fig. 3. Spatial distribution of measured data.

reducing the influence of variable dimensionality and model complexity on predictive performance [57–60].

The amount of water and cement influences the porosity and compactness of concrete structure by affecting the chemical bonding and physical adsorption capacity of cement paste and chloride ions. Existing results [61–63] indicated that the incorporation of mineral admixtures can effectively reduce temperature-induced cracks during early hydration. Therefore, it is necessary to study the influence of mineral admixtures type and content on the properties of marine concrete. Simultaneously, the influence of marine environment (such as chloride ion concentration in sea, temperature) and exposure time on the diffusion paths of chloride ions in concrete in different regions (splash, tidal and submerged zone) should be refined. Based on this, the input set established in this paper mainly covers three types of indicators: material components (water, cement content, fly ash content, slag content and silica fume content), regional characteristics (seawater chloride ion concentration, temperature) and time parameters (exposure time).

3.2. Data characterization

Based on the data and variable criteria established in Section 3.1, this paper established a data set of chloride ion concentration on the surface of concrete C_s and diffusion coefficient D containing 446 tidal zone [64–82], 131 splash zone [26,64,66,68,74,83–86], and 134 submerged zone [68–70,75,86–90] samples (Fig. 3). It should be noted that this paper uses the K nearest neighbor (KNN) to clean the abnormal data, which eliminates the obvious outliers by clustering the data.

The input set includes 8 indicators: cement content (OPC), fly ash content (FA), slag content (SG), silica fume content (SF), water content (Water), seawater chloride ion concentration (Cl^- in sea), seawater temperature (T), exposure time (t); the output set includes two indicators: chloride ion concentration on the surface of concrete C_s and chloride ion diffusion coefficient D. It should be noted that the database established in this paper was derived from the measured data of 11 seaports in Asia (Japan, China and Thailand), the Middle East (Persian Gulf) and Europe (Norway, Britain, Greece, etc.), so it can effectively analyze the influence of spatial and temporal distribution characteristics of different marine areas on the performance of marine concrete.

Considering the absence of specific values for the concrete chloride ion diffusion coefficient in some literature, so it is necessary to enhance the capacity of the database through data imputation [91,92]. The existing data imputation methods mainly include multiple statistics, regression, nearest neighbor interpolation and physical knowledge imputation. The first three algorithms interpolate missing data by quantifying sample similarity or sample probability distribution, while the fourth method employs expertise in the field to impute data and enhance the evolution patterns of data in real scenarios based on physical knowledge.

This paper used the physical knowledge method to interpolate the missing data. Specifically, the diffusion of chloride ions in concrete conforms to Fick 's second law (Eq. (4)), so the chloride ion diffusion coefficient can be deduced by determining the surface chloride ion concentration of concrete C_s and the chloride ion concentration $c(x,t)$.

$$c(x,t) = c_0 + (c_s - c_0)\left(1 - \frac{2}{\sqrt{\pi}}\int_0^{\frac{x}{2\sqrt{Dt}}} e^{-y^2}dy\right) \tag{4}$$

Where x is the distance from a point inside the concrete to the surface; t is the exposure time; D is the chloride ion diffusion coefficient of concrete; c_0 is the initial free chloride ion content in concrete;.

For the multi-training model, the distribution of high latitude data will significantly affect the model 's judgment of the output target [93–95]. Based on this, the frequency, correlation and statistical

Table 2
Distribution of characteristic variables in the database.

	OPC	FA	SG	SF	Water	Cl^- in sea	T	t	C_s	D
Min	110.00	0.00	0.00	0.00	38.50	0.08	7.00	13.00	0.02	0.10
Max	519.00	239.00	307.00	50.00	311.00	48.65	50.00	27.37	6.19	16.25
Mean	366.73	33.88	13.71	5.06	188.25	4.29	18.08	19.01	0.76	3.02
STD	79.85	60.01	49.78	12.45	42.89	6.41	9.08	2.72	0.64	2.43
Mode	340.00	0.00	0.00	0.00	180.00	5.00	7.00	19.00	0.28	14.96
Unit	kg/m^3	kg/m^3	kg/m^3	kg/m^3	kg/m^3	g/L	°C	year	%	10^{-12} m^2/s

Note: STD is the standard deviation

Table 3
Hyper-parameter optimization of multi-training models.

Model	Hyperparameters		
	Feature	Search space	Optimal value
GBDT	learning rate	[0.01–0.2]	0.05(0.1)
	max_depth	[2–5]	2(4)
	n_estimator	[50–200]	70(100)
XGB	learning rate	[0.01–0.2]	0.09(0.1)
	max_depth	[3–7]	5(4)
	n_estimator	[50–200]	100(200)
RF	min_samples_split	[2–10]	5(5)
	max_depth	[2–10]	10(5)
	min_samples_leaf	[1–4]	3(1)

Note: The values outside and inside the parentheses in the "Optimal value" represent the optimal parameter values for the predictive models of C_s and D, respectively.

characteristics (extremum, mean and standard deviation, etc.) of the data can be analyzed by explaining the distribution of each variable in the database. It can be found that the distribution ranges for Cl^- in seawater and T are 0.08–48.65 and 13–27.27 (Table 2), indicating that the database effectively covers the data distribution of environmental areas such as high temperature with high chloride salt. It is worth noting that the damage of concrete structures in low temperature environment is often caused by freeze-thaw cycles [96,97], which is not within the scope of analysis in this paper.

4. Results analysis and discussion

In this section, the typical tree models (RF, GBDT and XGB) were used to construct a multi-training model to simultaneously predict the surface chloride ion concentration and diffusion coefficient of concrete. Specifically, the grid search method was used to gradually optimize the common parameters of the tree models (Table 3), and then the 10 K-fold cross-validation was used to evaluate the prediction of the optimized models. It should be noted that 10 K-fold cross-validation is to randomly divide the dataset into 10 times. The proportion of the training set divided each time is 90 % and the test set is 10 %. The hyperparameters of the multi training model were shown in Table 3:

Fig. 4 compared the predicted and experimental values of surface chloride concentration of concrete by three tree models from the perspective of single sample. The proportion of data sample errors predicted by GBDT, RF and XGB models within 20 % is 69.8 %, 74.1 % and 68.4 %, respectively. It can be found that the ML models optimized by hyperparameters all show good prediction ability, and RF is the most suitable prediction model for surface chloride concentration of concrete. Some data points with larger prediction errors also reflect the influence of the discreteness of long term test data on the prediction performance of the models.

The prediction performance of three tree models for surface chloride concentration of concrete in different exposure zones was compared in Table 4. The prediction performance of the tree models in the tidal zone can be ranked as RF > GBDT > XGB. In this case, RF demonstrates the best prediction performance, with *RMSE*, R^2 and *MAE* are 0.15, 0.92 and 0.1, respectively. The prediction performance of the tree models in the splash zone can be ranked as RF > XGB > GBDT. RF also shows the best prediction performance, with *RMSE*, R^2 and *MAE* are 0.31, 0.89 and 0.18, respectively. It is because that the number of dry-wet cycles of concrete in the splash zone is unstable and significantly affected by climatic conditions, which leads to the large discreteness of the surface chloride ion concentration test data. Future efforts should focus on expanding the dataset from different maritime splash zones to enhance the predictive accuracy of ML models. The model prediction performance of RF and GBDT in the submerged area is similar. The *RMSE*, R^2 and *MAE* of the RF model are 0.17, 0.92 and 0.13, respectively, while the *RMSE*, R^2 and *MAE* of the GBDT model are 0.18, 0.91 and 0.13, respectively. Therefore, this paper chose RF as the prediction model of surface chloride concentration of concrete.

Fig. 5 compared the predicted and experimental values of chloride diffusion coefficient by three tree models from the perspective of single sample. The proportion of data sample errors predicted by GBDT, RF and XGB models within 20 % is 65.3 %, 68.9 % and 59.6 %, respectively. It can be found that RF is the most suitable prediction model for chloride diffusion coefficient. The main reasons for the large prediction error are as follows: (1) the discreteness of long-term test data; (2) some data are obtained by inverse calculation using Fick 's second law, which may not

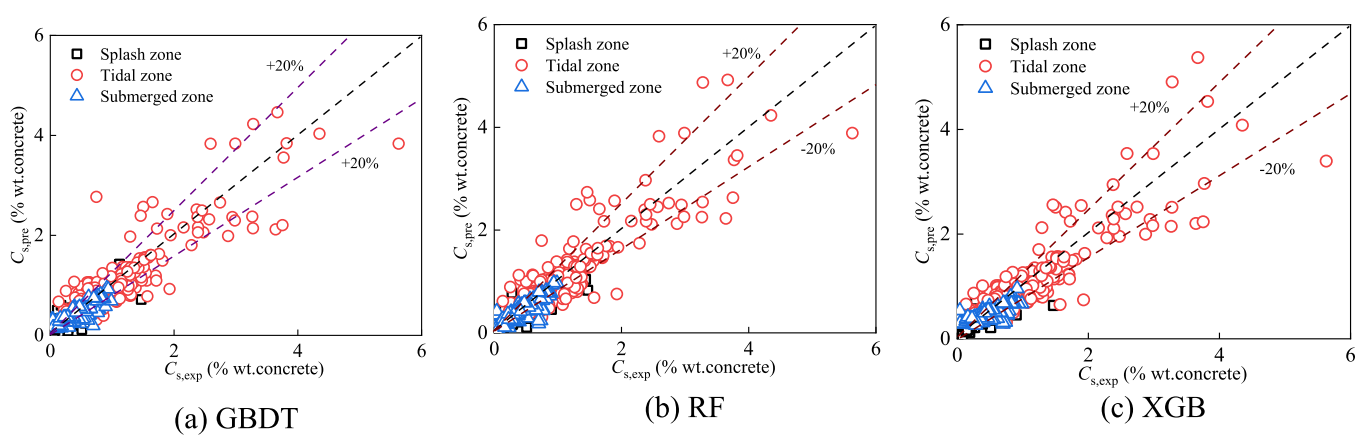

Fig. 4. Prediction results of surface chloride concentration of concrete.

Table 4
Prediction performance of multi training model on surface chloride concentration of concrete.

	Tidal zone			Splash zone			Submerged zone		
	GBDT	RF	XGB	GBDT	RF	XGB	GBDT	RF	XGB
RMSE	0.18	0.15	0.19	0.32	0.31	0.32	0.18	0.17	0.18
R^2	0.91	0.92	0.91	0.88	0.89	0.89	0.91	0.92	0.91
MAE	0.13	0.1	0.30	0.18	0.17	0.18	0.13	0.13	0.14

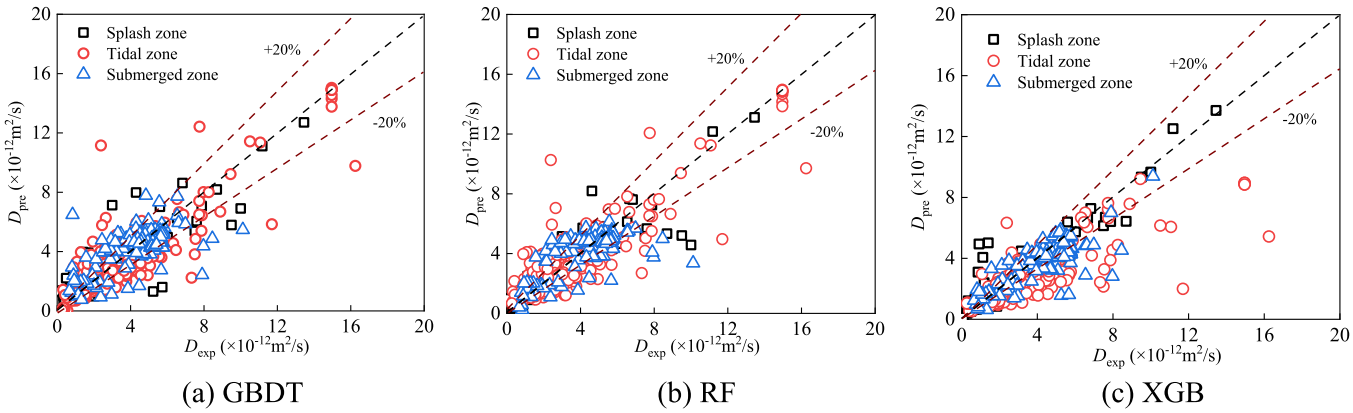

Fig. 5. Prediction results of chloride diffusion coefficient.

Table 5
Prediction performance of multi training model on chloride diffusion coefficient.

	Tidal zone			Splash zone			Submerged zone		
	GBDT	RF	XGB	GBDT	RF	XGB	GBDT	RF	XGB
RMSE	1.65	1.60	1.67	1.11	1.11	1.72	1.34	1.33	1.53
R^2	0.89	0.91	0.88	0.92	0.92	0.86	0.91	0.91	0.88
MAE	1.03	0.99	0.96	0.66	0.67	1.07	0.96	0.95	1.11

Table 6
Existing methods of surface chloride concentration of concrete.

No.	Design model	Equation	Marine environment
1	DuraCrete[14]	$C_s = A_c \cdot (w/b)$ where A_c is a coefficient related to the marine environment and the type of mineral admixture; w/b is water-to-binder ratio	Atmospheric; Splash; Tidal; Submerged
2	LNEC[98]	$C_s = 2.5(w/b) \cdot k_t \cdot C_b$; where k_t is a temperature-dependent coefficient; C_b is the chloride ion concentration related to coastal conditions	Atmospheric; Splash; Tidal; Submerged
3	Song et al.[99]	$C_s = 1.52\ln(3.77t + 1)$; where t is exposure time	Atmospheric; Splash; Tidal; Submerged
4	Chalee et al.[65]	$C_s = -[0.379(w/b) + 2.064]\ln(t) + 4.078(w/b) + 1.011$	Tidal
5	Petcherdchoo et al.[100]	$C_s = 10^{[0.814(w/b) - 0.213]} + 2.11t^{0.5}$	Tidal
6	Cai et al.[101]	$C_s = 10.01A_c \cdot (w/b) \cdot (1 - e^{-0.96t})$	Tidal; Splash
7	Yang et al.[102]	$C_s = 4.12A_c \cdot (w/b)\ C_{sw} \cdot (1 - e^{-0.56t})$; C_{sw} is the chloride concentration in sear	Submerged
8	Chen et al.[103]	$C_s = 9.4A_c \cdot (w/b) \cdot (5.49A_{ct} \cdot t)/(1 + 5.49A_{ct} \cdot t)$	Tidal
9	JTS 153–2015 [104]	$C_s = A_c \cdot (w/b)$	Atmospheric; Splash; Tidal

effectively reflect the influence of changes in experimental conditions and other parameters.

The prediction performance of three tree models for chloride diffusion coefficient in different exposure zones was compared in Table 5. The prediction performance of the tree models in the tidal zone can be ranked as RF > GBDT > XGB. In this case, RF demonstrates the best prediction performance, with *RMSE*, R^2 and *MAE* are 1.60, 0.91 and 0.99, respectively. The prediction performance of the tree models in the splash zone can be ranked as GBDT>RF>XGB. GBDT shows the best prediction performance, with *RMSE*, R^2 and *MAE* are 1.11, 0.92 and 0.67, respectively. The prediction performance of the tree models in the submerged zone can be ranked as RF>GBDT>XGB. RF also shows the best prediction performance, with *RMSE*, R^2 and *MAE* are 1.33, 0.91 and 0.95, respectively. Based on this, this paper used RF as the matrix model of multi training model.

5. Comparison of prediction accuracy of existing methods

This section analyzed the predictive accuracy of surface chloride concentration of concrete between 9 existing models (Table 6) and the multi-training models. Among them, the calculation method proposed by DuraCrete [14] and LNEC [98] analyzed the influence of different exposure zones on C_s by setting A_c. Song [99] proposed a prediction method by using the finite difference method. In addition, other scholars have proposed the calculation methods suitable for specific zones. It should be noted that the existing model used C_s (%wt. binder) as the calculation unit, while the model established in this paper used C_s (%wt. concrete) as the prediction basis. The prediction results of C_s can be converted by calculating the percentage of concrete mass and cementitious mass.

Fig. 6(a)-(c) compared the calculation results of various models in the splash zone, tidal zone, and submerged zone, respectively. The width of the blue boxes represents the quartile width of the predicted /experimental values, the line in the middle of the box represents the mean of the predicted /experimental values. Furthermore, the red data points and blue lines represent individual value and the distribution of the samples, respectively.

It can be found that the models with higher prediction accuracy in the tidal zone are DuraCrete and RF, with predicted mean values of 0.89 and 1.07 for surface chloride concentration of concrete, respectively. The prediction accuracy of the training model proposed in this paper is approximately 6.5 % higher than the second-ranked model. As shown in Fig. 6(b), except for JTS 153–2015, the other models exhibit better prediction accuracy in the splash zone. The prediction results obtained by the model established in this paper have the smallest discreteness (the smallest quartile). As shown in Fig. 6(c), the predicted mean values for the five models in the submerged zone are 1.13, 0.49, 3.3, 1.49, and 0.74, respectively.

In summary, the prediction model established in this paper can effectively predict the time dependent behavior of surface chloride

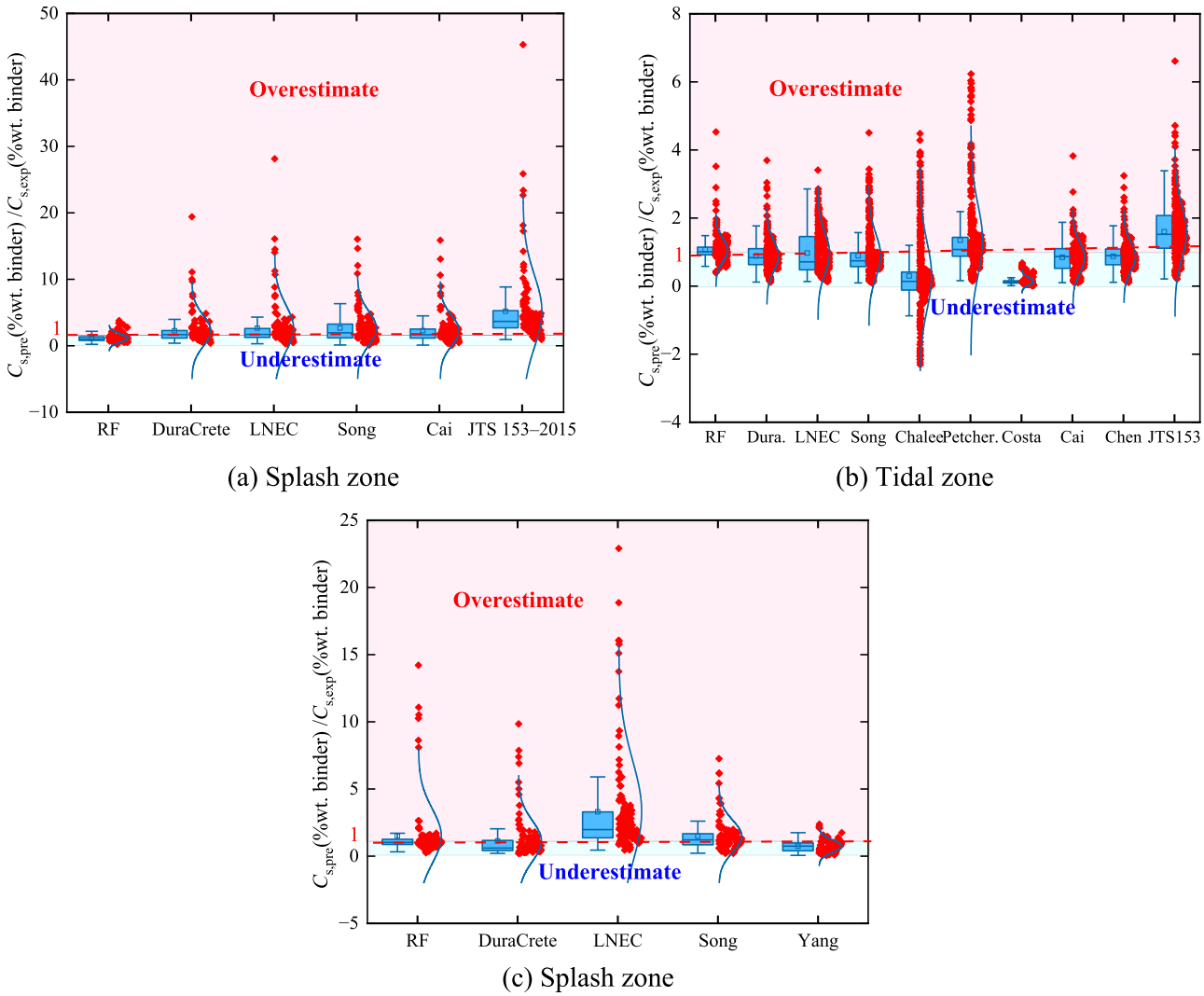

Fig. 6. Comparison of prediction accuracy between ML models and existing methods.

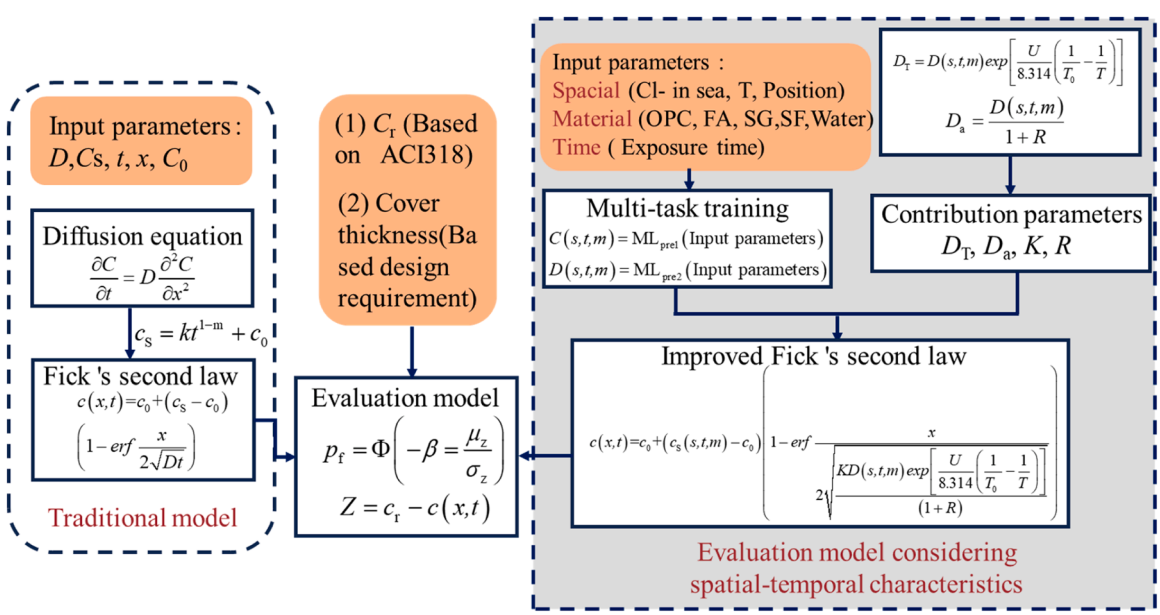

Fig. 7. Comparison of the proposed framework with traditional model.

concentration of concrete in different zones, which also reflects the importance of multivariate processing capacity for concrete durability assessment.

6. Service life evaluation model of marine concrete structures

6.1. Establishment of the model

Under the coupled effects of environmental conditions and loads, the generation of microcracks and defects within concrete will accelerate the diffusion of chloride ions internally, resulting in a significant reduction in the service life of marine concrete structures [105–107].

Existing chloride ion transport model for marine concrete structures mainly used Fick 's second law (Fig. 7). This method is based on the diffusion theorem and has the following disadvantages: (1) Ignore the degradation of chloride ion diffusion coefficient over exposure time; (2) Neglect the bonding capacity between concrete and chloride ions; (3) Overlook the influence of parameters such as seawater temperature, Cl^- in sea, and material composition in different zones on the structural life.

To address these issues, this paper proposed a service life evaluation model of marine concrete structures that can consider the above factors. Specifically, the multi-training model established in Section 4 is used to predict the surface chloride concentration $C_s(s,t,m)$ and chloride diffusion coefficient $D(s,t,m)$. Thus, $C_s(s,t,m)$=ML_{pre1}(input parameter) and $D(s,t,m)$=ML_{pre2} (input parameter). Furthermore, there are two states of chloride ion into concrete: (1) Free chloride ions in the pore solution; (2) Binding chloride ions adsorbed on concrete. Considering that the free chloride ion content is the key factor to the corrosion of steel, this paper used the conversion method proposed by Nilsson et al. [108] to calculate the free chloride ion content (Eq. (5)). And the calculation method proposed by Martin-Perez B et al. [109] (Eq. (6)) was used to analyze the effect of temperature on the chloride ion diffusion coefficient:

$$D_a = \frac{D(s,t,m)}{1+R} \tag{5}$$

$$D_T = D(s,t,m)exp\left[\frac{U}{8.314}\left(\frac{1}{T_0}-\frac{1}{T}\right)\right] \tag{6}$$

Where D_a is the free chloride ion content; D_T is the surface chloride ion diffusion coefficient of concrete at temperature T; T_0 is the temperature at 28 day; U is the activation energy for chloride ion diffusion, DuraCrete [14] recommended value is 39.9 kJ / mol; R represents the binding capacity between chloride ions and concrete, with a recommended value of 0.36 according to Ref [110].

Furthermore, this paper used the deterioration coefficient K to reflect the ratio of chloride ion diffusion coefficient of concrete under service and experimental environment, and then substituted Eq. (5)-(6) into Eq. (4) to obtain the improved chloride ion transport model proposed in this paper (Eq. (7)):

$$c(x,t) = c_0 + (c_S(s,t,m) - c_0)\left(1 - erf\frac{x}{2\sqrt{\frac{KD(s,t,m)exp\left[\frac{U}{8.314}\left(\frac{1}{T_0}-\frac{1}{T}\right)\right]}{(1+R)}}}\right) \tag{7}$$

The service life of marine concrete structures depends on the probability of these structure fulfilling the expected function in the exposure environment. Accordingly, this paper established a reliability model to evaluate the service life of the structures. Firstly, the model resistance is determined by the critical chloride ion content c_r for steel corrosion, and the loading effect is the concentration of free chloride ions on the surface of steel $c(x_k,t)$. Therefore, the performance function of the model Z = G (c_r, $c(x_k,t)$)= c_r-$c(x_k,t)$. And the failure probability P_f of the concrete structure can be expressed as:

Table 7
Probability distribution of uncertain parameters.

Variables	Mean	Variance	Distribution	Probability
x_k	7	1.4	Normal	N~(7,1.4)
R	0.38	0.07	Normal	N~(0.38,0.07)
K	1.81	0.36	Normal	N~(1.81,0.36)

$$P_f = \int_{-\infty}^{-\frac{\mu_z}{\sigma_z}} \varphi(X)dX = \Phi\left(-\frac{\mu_z}{\sigma_z}\right) = \Phi(-\beta) = 1 - \Phi(\beta) \tag{8}$$

6.2. Parameter sensitivity and uncertainty quantification

This paper simulated the failure probability of concrete structures in the Qingdao wheat island, China. Then the influence of uncertain parameters such as concrete cover thickness x_k, chloride binding capacity R and deterioration coefficient K on the service life of structure was quantified. It should be noted that the material composition of concrete and environmental parameters adopted the measured data provided in Ref [76]. The critical chloride ion content c_r used the values recommended by DuraCrete [14] (tidal zone is selected as the exposed zone), the temperature (T_0) is 10 °C, the cover thickness (x_k), chloride ion binding capacity (R), and degradation factor (K) all followed the probability distribution model recommended by Wu et al. [111] (Table 7).

Based on probability and statistical theory, MCS evaluates the safety of structures by randomly sampling the probability distribution of given variables [112–114]. Due to its capabilities in high dimensional variable analysis, nonlinear limit function solving, and rapid convergence, MCS has been extensively applied in the reliability analysis of concrete structures [115–118]. Based on this, this paper used MCS to randomly sample the uncertain parameters. It is worth noting that the high-precision evaluation results of Monte Carlo depend on the number of simulations for generating samples, but this will also lead to computational memory and time-related issues. This paper used the method proposed by Ref [119] to analyze the sampling number N from the perspective of failure probability P_f, that is, N ≥ 100/P_f. Therefore, the number of simulations N = 10000 > 100/5 %= 2000 was set in this paper. In summary, the framework of sensitivity analysis and uncertainty quantification established in this paper is as follows: (1) Calculate the deterministic parameters in Eq. (7), such as C_s and D (from ML models), c_r (from DuraCrete), T_0 = 10 °C and T (from the input parameters); establish the probability distribution of uncertain parameters and generate 10,000 random samples using Monte Carlo; Incorporate the parameters into the service life model of marine concrete structures established in this paper; (2) Quantify the variation patterns of the structural failure probability with service time, concrete cover thickness, and slag content, etc. In order to reduce the influence of sampling randomness on P_f, 10 simulations are carried out and the average value is taken as the P_f. (3) Establish various combinations of uncertain and deterministic parameters to quantify the influence of uncertain parameters on the structural failure probability.

Fig. 8(a)-(d) analyzed the influence of cover thickness x_k, slag content r_{slag}, T and Cl^- in sea on the failure probability of the structure. Generally, P_f increases with the increase of the service time. It can be attributed to the accumulation of free chloride ions on the surface of the steel over the service time, ultimately leading to structural deterioration.

However, P_f will decrease with the increase of x_k (Fig. 8(a)). When x_k is 6,6.5,7 and 7.5 cm, the service life of the structure (P_f =5 %) is 49.5, 55.4, 59.8 and 62.4 years, respectively. This is because the enlargement of the chloride ion diffusion path with the increase of x_k, thereby reducing the probability of steel corrosion. Compared with x_k of 6 cm, the service life of structures with x_k of 6.5, 7, and 7.5 cm increases by 11.9 %, 19.9 %, and 24.2 %, indicating that the growth rate of the service life decreases with the increase of x_k.

Fig. 8(b) showed that P_f will decrease with the increase of r_{slag}. When

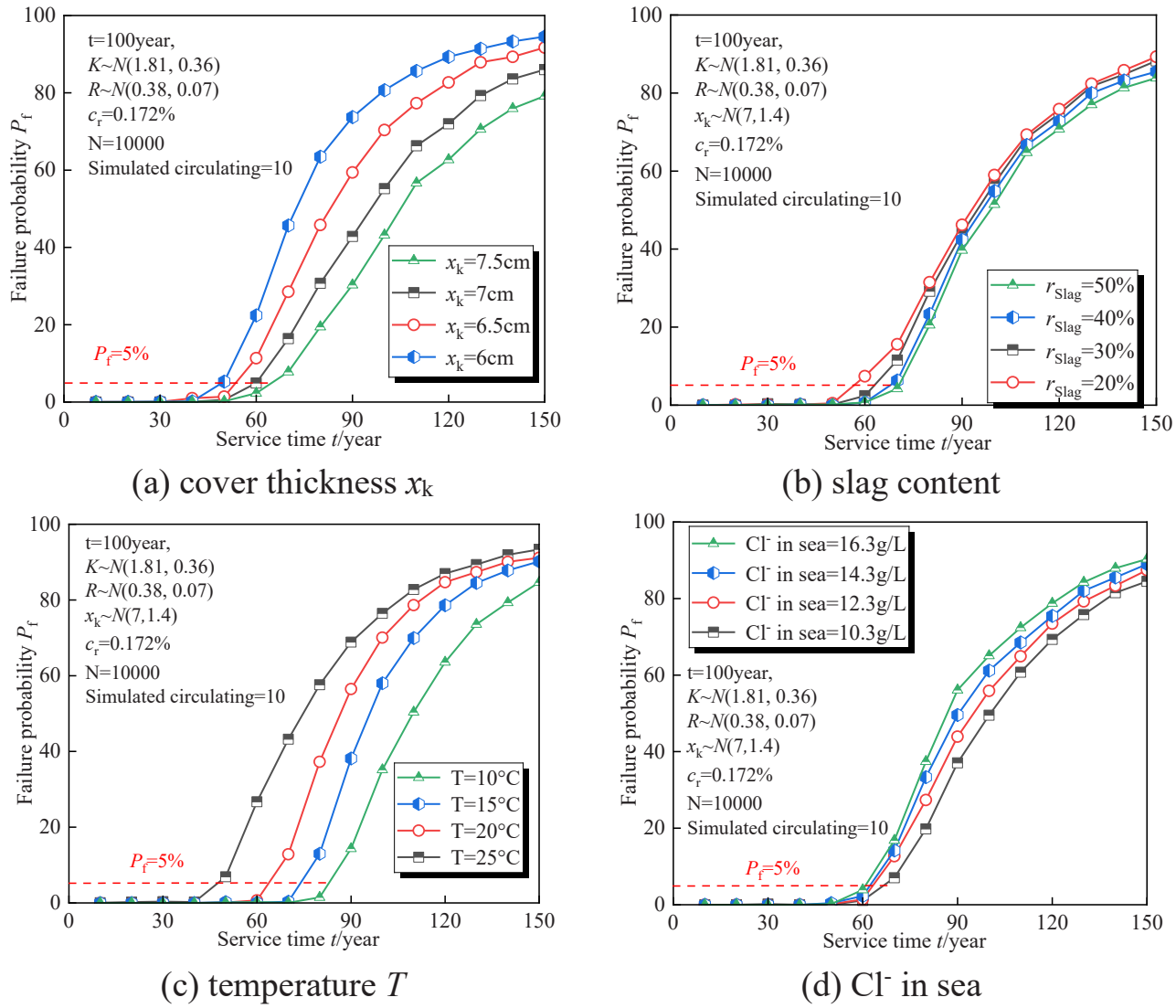

Fig. 8. Influence of characteristic parameters on P_f.

r_{slag} is 20,30,40 % and 50 %, the service life (P_f =5 %) is 56.2,63.5,68.2 and 71 years, respectively. Compared with r_{slag} is 20 %, the service life with r_{slag} of 30,40 % and 50 % increased by 13.0 %, 20.4 % and 24.5 %, respectively. This is because the incorporation of slag will enhance the compactness and durability of the concrete structure, thereby reducing the probability of structural failure.

Fig. 8(c) showed that P_f increases with the increase of T. When T is 10, 15, 20 and 25 °C, the service life (P_f =5 %) is 83.5, 74.2, 63.8 and 48.5 years, respectively. Compared with T of 10 °C, the service life of at T of 15, 20 and 25 °C decreases by 11.1 %, 23.5 % and 47.8 %, respectively. This is because the activation energy of chloride ion diffusion in concrete will increase with the increase of T,

thereby enhancing the ability of chloride ions to diffuse to the steel.

Fig. 8(d) showed that P_f increases with the increase of Cl^- in sea. When Cl^- in sea is 10.3, 12.3, 14.3 and 16.3 g/L, the service life (P_f =5 %) is 67.7, 63.1, 62.4 and 60.9 years, respectively. Compared with Cl^- in sea of 10.3 g/L, the service life of at Cl^- in sea of 12.3, 14.3 and 16.3 g/L decreases by 5.4 %, 6.5 % and 8.9 %, respectively. With the increase of Cl^- in sea concentration, the chloride ion concentration of concrete on the surface of steel bar also increases, which reduces the reliability of the structure.

Fig. 9(a)-(c) analyzed the influence of uncertain parameter combinations on P_f. Fig. 9(a) set R as an uncertain parameter, and x_k and K were set as deterministic parameters with values of 7 and 1.81 to evaluate the influence of single probability distribution on P_f. Fig. 9(b) set K and R as uncertain parameters, and x_k was set as a deterministic parameter with values of 7 to evaluate the influence of two probability distribution on P_f. In Fig. 9(c), K, R and x_k were set as uncertain parameters to evaluate the influence of multiple probability distributions on P_f. The remaining deterministic parameters used the measured data recommended by Ref [76]. It should be noted that the blue line in the figure represents the cumulative distribution function of the uncertain variables, illustrating the cumulative distribution of samples.

It can be found that P_f obeys the normal distribution $N\sim(75.9,1.6\times10^{-2})$, When the single probability distribution model is used. When employing a two-probability distribution model, P_f obeys the normal distribution $N\sim(63.8,2.2\times10^{-2})$. Furthermore, with the adoption of a three-probability distribution model, P_f obeys the normal distribution $N\sim(55.1,5.29\times10^{-2})$. Compared with the single probability distribution, the P_f of the two-probability and three-probability distribution model decrease by 15.9 % and 29.6 %, respectively. The results showed the number of uncertain variables will significantly increase the complexity of the model, and then affect the probability distribution of P_f.

It can be observed that as the sample size N approaches infinity, the probability distribution of P_f tends to a normal distribution, which is also

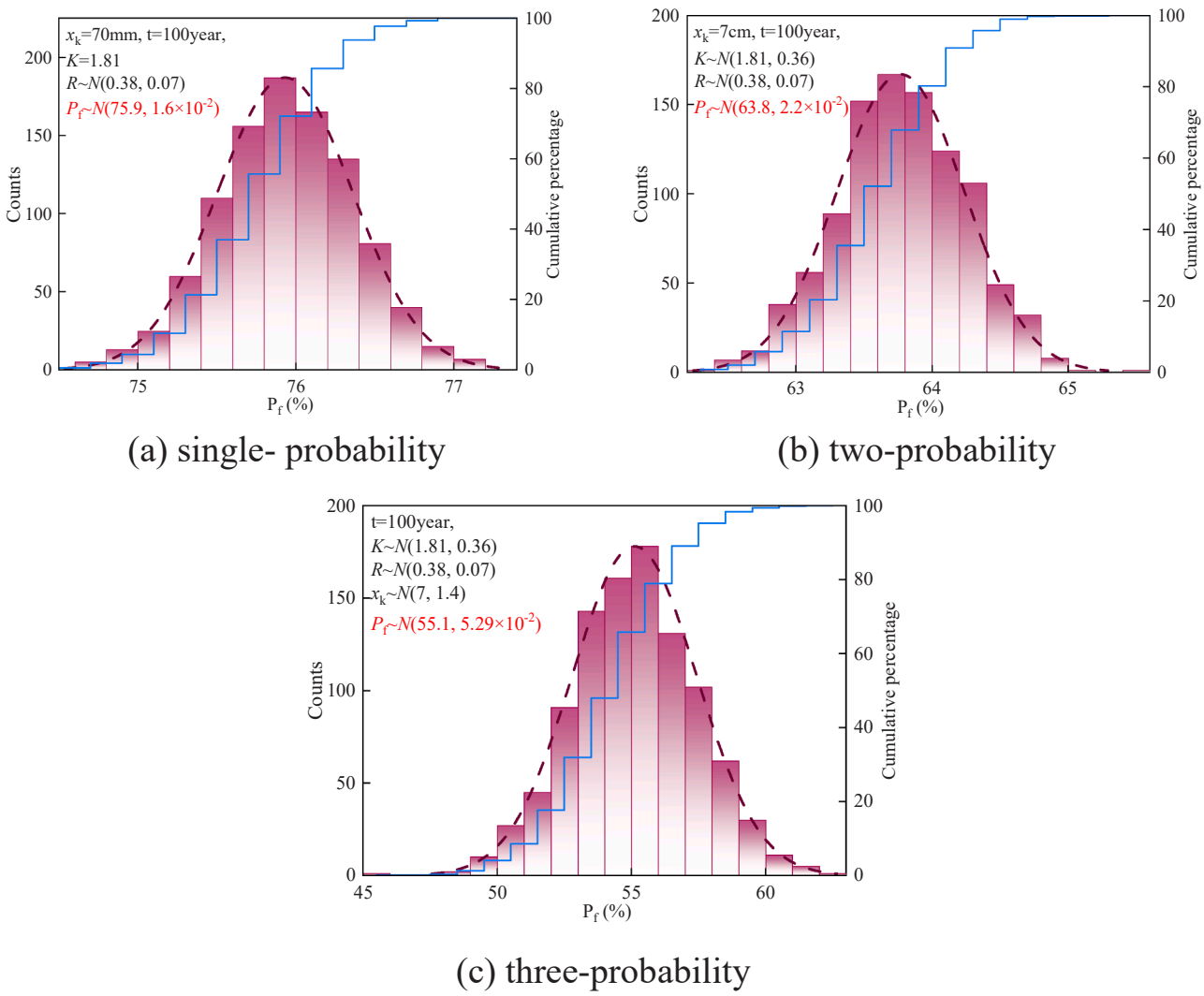

Fig. 9. Quantization of uncertain parameters.

consistent with the central limit theorem. This is because the uncertainty parameters in this paper obey the assumption of independent and identical distribution.

It is worth noting that there is no measured data on the deterioration coefficient K and chloride binding capacity R. Consequently, specific probability distributions for the mentioned uncertainty parameters and their impacts on the service life of structures cannot be determined. In the future, it is essential to conduct relevant experiments and establish corresponding databases, and then the probability distribution of uncertain parameters can be determined by Bayesian reasoning and Markov chain, so as to evaluate the influence of such parameters on the service life of structures.

7. Optimization of cover thickness

Existing studies have shown that the diffusion path of chloride ions and the time required to reach the surface of the steel will increase with increase of the cover thickness of concrete, thereby reducing the probability of steel corrosion during the service time [120–122]. when the cover thickness is too large, the growth rate of the service life will gradually decrease with the increase of the cover thickness. In addition, the shrinkage and creep of concrete will lead to cracking in the tensile zone of the structure, thereby reducing the beneficial effect of concrete on steel.

Based on this, an optimization model of concrete cover thickness based on NLS was established in this paper. The model can provide the optimal design value of concrete cover thickness for various parameter combinations, including material composition, regional spatiotemporal characteristics, and reliability indicators. Specifically, different proportions of OPC and SG were set to measure the impact of material compositions on the cover thickness. Different temperatures, Cl^- in sea and service time were set to simulate the influence of the regional environment. The failure probability used the values recommended by GB50068–2018 [123] to evaluate the influence of design requirements on the cover thickness. The specific values of parameters were as follow: OPC= 270–360 kg/m^3, SG= 90–180 kg/m^3 (r_{Slag}=20 %–40 %), Cl^- in sea= 12.3–16.3g/L, T = 15–25 °C, P_f= 5 % or 10 %. In addition, the optimization model can also effectively analyze the influence of SF and FA on the cover thickness by setting different SF and FA contents.

It can be found that the optimization of concrete cover thickness under multi-parameter combination is essentially a nonlinear optimization problem. Therefore, this paper used nonlinear least squares method to solve the above problems. Firstly, the optimization objective is to find the cover thickness that minimizes the residual sum of square (RSS) between designed reliability indicators and calculated values (Eq. (9)). The commonly method for minimizing RSS is to differentiate the cover thickness and set the equation to 0 (Eq. (10)). Subsequently, the solution to this equation can be obtained:

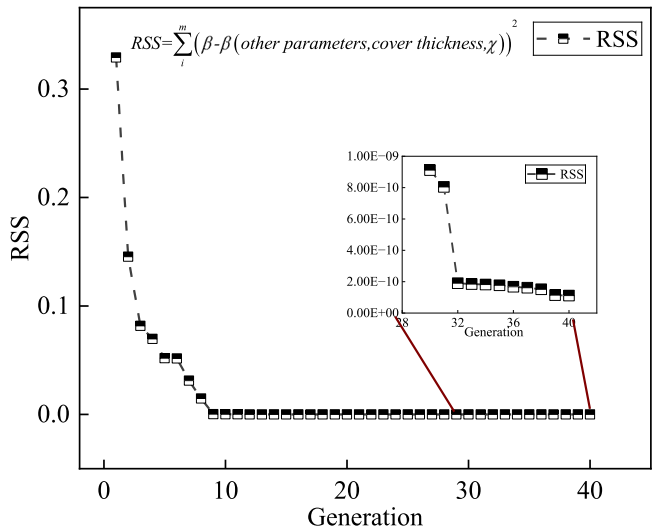

Fig. 10. Evolution of RSS with the number of iteration.

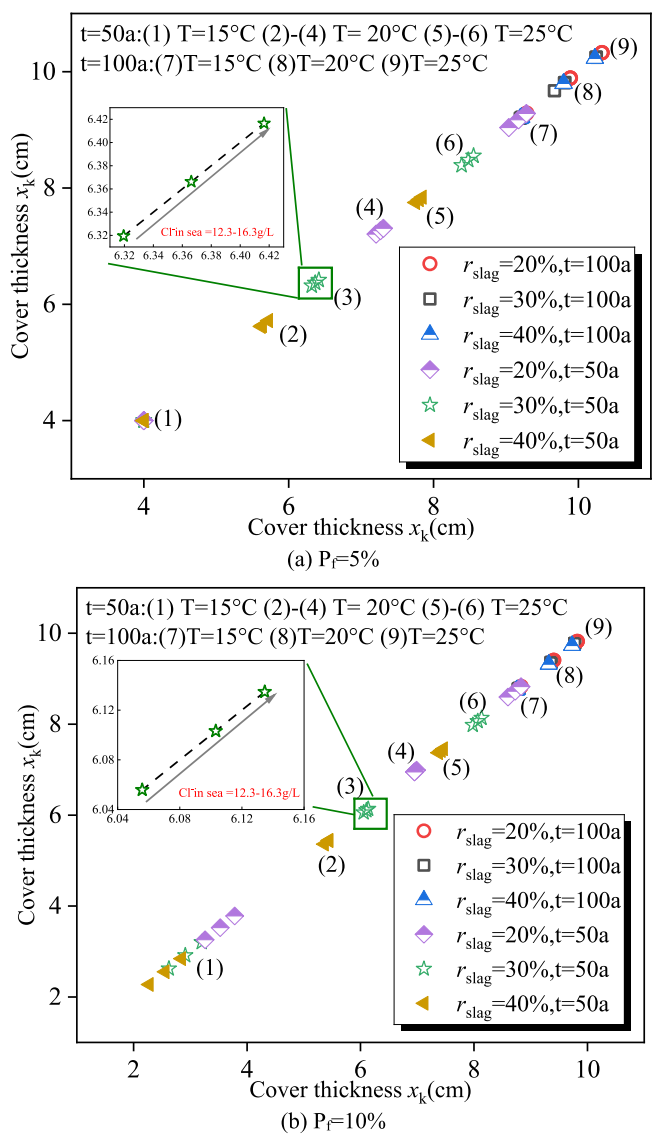

Fig. 11. Distribution of concrete cover thickness under different working conditions.

$$RSS = \sum_{i}^{m} (\beta - \beta(others, x_k, \chi))^2 \tag{9}$$

$$\frac{\partial}{\partial(x_k)} RSS = 0, \forall j \in [1, others] \tag{10}$$

$$\frac{\partial}{\partial(x_k)} RSS = -2\sum_{i}^{m} (\beta - \beta(others, x_k, \chi)) \cdot \frac{\partial\beta(others, x_k, \chi)}{\partial(x_k)} = 0 \tag{11}$$

Where *others* are the other parameters of the model; χ is the coefficient of T, R and *others* in the life evaluation model.

It can be found that the system of partial differential equations has optimization parameter (cover thickness), known parameters (c_r, Cl^- in sea, R, etc.) and χ. So the number of solved equations is a total of known parameters multiplied by parameter combinations, which is computationally challenging and easily leads to the disappearance of the optimization gradient.

Based on this, the Levenberg-Marquardt (LM) algorithm was used to solve the optimization model, which has advantages of the gradient method and the newton method. The specific process is as follows: (1) Initialize the value of the optimization parameter (x_k), which can be determined by engineering experience; (2) Calculate the Jacobi matrix J of *RSS*; (3) Set the initial damping factor λ of LM, the maximum number of iterations, the iteration accuracy and size of each step; (4) Calculate the updated value of x_k, the calculation method is shown in Eqs. (12) - (13); (5) Substitute the updated x_k into Eq. (9) and calculate *RSS*. When the value of *RSS* is less than the iteration accuracy, the calculation of the model is terminated:

$$\Delta x_k = -\left(J^T J + \lambda I\right)^{-1} J^T RSS \tag{12}$$

$$x_{k,i+1} = x_{k,i} + \Delta x_k \tag{13}$$

Where I is the unit matrix.

It can be found that the optimization model established in this paper can effectively predict the cover thickness of concrete under different parameter combinations (Fig. 10). After 40 iterations, the prediction error *RSS* decreases from 0.33 to 1.61×10^{-10}.

In addition, Fig. 11 analyzed the distribution of x_k corresponding to 108 working conditions. When $P_f = 5$ % and $t = 100$ year (Fig. 11(a)), the cover thickness corresponding to 27 working conditions ranges from 9.10 to 10.32 cm. When x_k is 10.32 cm, the most unfavorable working condition is $T = 25$ °C + Cl^-in sea= 16.3 g/L + r_{slag}= 20 %. When x_k is 9.10 cm, the corresponding optimal working condition is $T = 15$ °C + Cl^-in sea= 12.3 g/L + r_{slag}= 40 %. Compared with the optimal condition, the x_k of the unfavorable condition needs to be increased by 13.4 %. When $P_f = 5$ % and $t = 50$ years, the x_k corresponding to the 27 working conditions ranges from 3.98 to 9.28 cm, and the most unfavorable condition and the optimal condition are the same as $t = 100$ years. In this case, the x_k of the most unfavorable condition needs to be increased by 138.5 %.

When $P_f = 10$ % and $t = 100$ year (Fig. 11(b)), the x_k corresponding to 27 working conditions ranges from 8.74 to 9.82 cm. In this case, the x_k of the unfavorable condition needs to be increased by 12.4 %. When $P_f = 10$ % and $t = 50$ years, the x_k corresponding to the 27 working conditions ranges from 2.27 to 8.81 cm. In this case, the x_k of the unfavorable condition needs to be increased by 288.5 %. It should be noted that the most unfavorable conditions and optimal conditions corresponding to $P_f = 10$ % are the same as those of $P_f = 5$ %.

It can be found that multi parameter combinations have significantly influence on the x_k ($t = 50$ year). This is because the failure rate of the structure in the early stage is controlled by the steel depassivation caused by chloride ion diffusion. The slag content, temperature and other indicators are the key variables affecting the chloride ion diffusion capacity of concrete. As the service time increases, the chloride ion diffusion coefficient of concrete decreases continuously and the steel

Table 8
The cover thickness of concrete under different parameter combinations.

x_k (cm)	Parameter combinations (P_f=5 %)	Parameter combinations (P_f=10 %)
6	t = 50year + T = 15 −20 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 40 %; t = 50year + T = 15 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 20 −30 %;	t = 50year + T = 20 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 40 %; t = 50year + T = 15 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 20 −40 %;
7	t = 50year + T = 20 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 30 %	t = 50year + T = 20 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 20 −30 %
8	t = 50year + T = 25 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 40 %; t = 50year + T = 20 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 20 %;	t = 50year + T = 25 °C + Cl^-in sea = 12.3 g/L + r_{slag}= 30 %; t = 50year + T = 25 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 40 %
9	t = 50year + T = 25 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 30 %	t = 100year + T = 15 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 20 −40 %; t = 50year + T = 25 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 20 %; t = 50year + T = 25 °C + Cl^-in sea = 14.3 −16.3 g/L + r_{slag}= 30 %
10	t = 100year + T = 15 −25 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 20 −40 %; t = 50year + T = 25 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 20 %	t = 100year + T = 20 −25 °C + Cl^-in sea = 12.3 −16.3 g/L + r_{slag}= 20 −40 %

depassivation is not the main mode of structural failure. Therefore, the multi-parameter combination has little effect on the x_k (t = 100 year).

In the parameter combinations established in this study, the change of x_k caused by other factors except T is less than 1 cm. Therefore, 108 parameter combinations can be divided according to the value distribution of x_k to improve the efficiency of engineering design (Table 8). For example, when P_f= 5 % and x_k = 6 cm, the corresponding design conditions are as follows: (1) t = 50year + T = 15–20 °C + Cl^-in sea = 12.3–16.3 g/L + r_{slag}= 40 %; (2) t = 50year + T = 15 °C + Cl^-in sea = 12.3–16.3 g/L + r_{slag}= 20–30 %. The results showed that the optimization model established in this paper can be used to simulate the design of concrete cover thickness under any multi-parameter combination. It should be noted that when the optimal values x_k < 6 cm or x_k > 10 cm, the x_k can be taken as 6 cm and 10 cm to avoid structural performance degradation caused by too small or too large cover thickness.

8. Conclusion

This paper established an intelligent service life evaluation framework for marine concrete structures. Firstly, a database containing 711 sets of surface chloride ion concentrations and diffusion coefficients in different exposure zones was constructed, and tree models optimized by the grid search method were used as the multi training matrix to predict the above variables. An evaluation model that can comprehensively consider multi-factors was established. The influence of uncertain parameters on the failure probability was quantified. The NLS-LM method was used to determine the cover thickness under multi-parameter combinations. The main conclusions are as follows:

(1) The multi-training models optimized based on grid search demonstrate effective predictions for surface chloride ion concentrations and diffusion coefficients of concrete. RF exhibits the highest predictive accuracy, with mean prediction values of 1.07, 1.11, and 1.13 in the splash zone, tidal zone, and submerged zone, respectively.
(2) The evaluation model based on multi training and Monte Carlo simulation can effectively consider the influence of regional spatial and temporal characteristics, material composition and other factors. The failure probability of the structure will decrease with the increase of cover thickness and slag content, and increase with the increase of chloride ion concentration and seawater temperature.
(3) The probability distribution and combination of uncertain parameters significantly affect the failure probability of the structure. When the single, two and three probability distribution models are used, the failure probability of the structure follows the normal distribution N~(75.9,1.6 ×10^{-2}), N~(63.8,2.2 ×10^{-2}) and N~(55.1,5.29×10^{-2}), respectively.
(4) The optimization model based on NLS-LM algorithm can simulate the design of cover thickness under any multi-parameter combination. When P_f= 5 % and t = 100year, the recommended value of cover thickness is 10.32 cm under the most unfavorable condition (T = 25 °C + Cl^-in sea=16.3 g/L + r_{slag}=20 %).

CRediT authorship contribution statement

Shiqi Wang: Writing – original draft, Software, Resources. **Renjie Wu:** Writing – original draft, Validation, Supervision. **Fuyuan Gong:** Writing – review & editing, Validation, Funding acquisition. **Jin Xia:** Validation, Supervision. **Yuxi Zhao:** Writing – review & editing, Supervision. **Bin Zeng:** Validation, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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